# ClipSAT : Math for AI & Data Science

## Unit 7.01: Logistic Regression & Classification

Name: _____

**Q1.** In evaluating a binary classifier, the confusion matrix shows TP = 45 , TN = 40 , FP = 10 , and FN = 10 . Compute the overall classification accuracy.

[ ] A: 81.0%
[ ] B: 81.8%
[ ] C: 52.4%
[ ] D: 13

**Q2.** In evaluating a binary classifier, the confusion matrix shows TP = 55 , TN = 40 , FP = 10 , and FN = 10 . Compute the overall classification accuracy.

[ ] A: 82.6%
[ ] B: 84.6%
[ ] C: 56.5%
[ ] D: 13

**Q3.** In evaluating a binary classifier, the confusion matrix shows TP = 60 , TN = 40 , FP = 10 , and FN = 10 . Compute the overall classification accuracy.

[ ] A: 83.3%
[ ] B: 85.7%
[ ] C: 58.3%
[ ] D: 13

**Q4.** In evaluating a binary classifier, the confusion matrix shows TP = 70 , TN = 40 , FP = 10 , and FN = 10 . Compute the overall classification accuracy.

[ ] A: 84.6%
[ ] B: 87.5%
[ ] C: 61.5%
[ ] D: 13

**Q5.** A spam classification model identifies TP = 30 genuine spam emails and FP = 10 legitimate emails falsely flagged as spam. What is the precision of this model?

[ ] A: 75.0%
[ ] B: 60.0%
[ ] C: 25.0%
[ ] D: 85.7%

**Q6.** A spam classification model identifies TP = 60 genuine spam emails and FP = 20 legitimate emails falsely flagged as spam. What is the precision of this model?

[ ] A: 75.0%
[ ] B: 60.0%
[ ] C: 25.0%
[ ] D: 85.7%

**Q7.** A spam classification model identifies TP = 90 genuine spam emails and FP = 30 legitimate emails falsely flagged as spam. What is the precision of this model?

[ ] A: 75.0%
[ ] B: 60.0%
[ ] C: 25.0%
[ ] D: 85.7%

**Q8.** A spam classification model identifies TP = 120 genuine spam emails and FP = 40 legitimate emails falsely flagged as spam. What is the precision of this model?

[ ] A: 75.0%
[ ] B: 60.0%
[ ] C: 25.0%
[ ] D: 85.7%